

11.24

```
library(AER)
library(car)
```

11.24 Reconsider Example 11.2 on the supply and demand for fish at the Fulton Fish Market. The data are in the file fultonfish.

```
load("/Users/liaochenghao/Desktop/Financial Econometrics/Homework/HW7/Data/fultonfish.rdata")
df <- fultonfish
names(df)
```

```
## [1] "date" "lprice" "quan" "lquan" "mon" "tue" "wed" "thu"
## [9] "stormy" "mixed" "rainy" "cold" "totr" "diff" "change"
```

- Add the variable MIXED, which indicates poor but not STORMY weather conditions, to the supply equation in equation (11.14). Estimate the new reduced-form equation for $\ln(\text{PRICE})$, adding the variable MIXED to equation (11.16). Is it statistically significant at the 5% level? Test the joint significance of STORMY and MIXED. Is the resulting F-value greater than 10?

$$\ln(\text{QUAN}_t) = \beta_1 + \beta_2 \ln(\text{PRICE}_t) + \beta_3 \text{STORMY}_t + e_{st} \quad (11.14)$$

$$\ln(\text{PRICE}_t) = \pi_{12} + \pi_{22} \text{MON}_t + \pi_{32} \text{TUE}_t + \pi_{42} \text{WED}_t + \pi_{52} \text{THU}_t + \pi_{62} * \text{STORMY}_t + v_{t2} \quad (11.16)$$

```
# estimate new reduced-form equation
new_rf_model <- lm(lprice ~ mon + tue + wed + thu + stormy + mixed, data = df)
# Test the significance of MIXED.
summary(new_rf_model)$coefficients["mixed", "Pr(>|t|)"]
```

```
## [1] 0.003209729
```

```
# Test the joint significance of STORMY and MIXED.
linearHypothesis(new_rf_model, c("stormy = 0", "mixed = 0"))
```

```
##
## Linear hypothesis test:
## stormy = 0
## mixed = 0
##
## Model 1: restricted model
## Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed
##
```

```
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     106 15.876
## 2     104 12.115  2    3.7604 16.14 7.857e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

p-value ≤ 0.05 , reject H_0 Yes, it is statistically significant at the 5% level?

Yes, the resulting F-value is greater than 10.

b. Estimate the demand equation using STORMY and MIXED as IVs. Compare the coefficient estimates to those in Table 11.5.

demand equation:

$$\ln(QUAN_t) = \alpha_1 + \alpha_2 \ln(PRICE_t) + \alpha_3 MON_t + \alpha_4 TUE_t + \alpha_5 WED_t + \alpha_6 THU_t + e_{dt} \quad (11.13)$$

```
iv_model <- ivreg(
  lquan ~ lprice + mon + tue + wed + thu |
  mon + tue + wed + thu + stormy + mixed,
  data = df
)

summary(iv_model)

##
## Call:
## ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
##       tue + wed + thu + stormy + mixed, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1327 -0.4320  0.0654  0.5133  1.1798
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   8.54023    0.15661  54.533 < 2e-16 ***
## lprice       -0.93014    0.35340  -2.632  0.00977 **
## mon          -0.01190    0.20837  -0.057  0.95456
## tue          -0.52583    0.20230  -2.599  0.01068 *
## wed          -0.56262    0.20696  -2.719  0.00767 **
## thu           0.09987    0.20285   0.492  0.62350
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6853 on 105 degrees of freedom
## Multiple R-Squared:  0.185,    Adjusted R-squared:  0.1462
## Wald test: 4.927 on 5 and 105 DF, p-value: 0.0004317
```

Using STORMY and MIXED as instruments for $\ln(PRICE)$, the 2SLS estimate of the demand equation gives a coefficient on $\ln(PRICE)$ of -0.9301, with a standard error of 0.3534 and a p-value of 0.00977. Therefore, the price coefficient is statistically significant at the 5% level.

Compared with Table 11.5, where the coefficient on $\ln(\text{PRICE})$ is -1.1194, the new estimate is smaller in absolute value. However, the sign remains negative, which is consistent with the law of demand. The estimated day-of-week effects are also very similar to those in Table 11.5. Therefore, adding MIXED as an additional instrument changes the magnitude of the price elasticity somewhat, but the overall conclusion remains the same: fish demand decreases as price increases.

c. Test the validity of the surplus instrument using the Sargan test, discussed in Section 10.3.4.

```
summary(iv_model, diagnostics = TRUE)

##
## Call:
## ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
##       tue + wed + thu + stormy + mixed, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1327 -0.4320  0.0654  0.5133  1.1798
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.54023     0.15661  54.533 < 2e-16 ***
## lprice       -0.93014     0.35340  -2.632  0.00977 **
## mon          -0.01190     0.20837  -0.057  0.95456
## tue          -0.52583     0.20230  -2.599  0.01068 *
## wed          -0.56262     0.20696  -2.719  0.00767 **
## thu           0.09987     0.20285   0.492  0.62350
##
## Diagnostic tests:
##              df1 df2 statistic  p-value
## Weak instruments    2 104    16.140 7.86e-07 ***
## Wu-Hausman          1 104     1.489   0.225
## Sargan              1  NA     0.772   0.380
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6853 on 105 degrees of freedom
## Multiple R-Squared:  0.185,    Adjusted R-squared:  0.1462
## Wald test: 4.927 on 5 and 105 DF,  p-value: 0.0004317
```

Sargan test p-value = 0.380 > 0.05, reject H_0 : The instruments are valid

d. In the reduced-form equation in part (a), test the joint significance of the indicator variables MON, TUE, WED, and THU at the 5% level. What do you conclude? Are we now able to estimate the supply equation by 2SLS with confidence in our procedure?

```
linearHypothesis(
  new_rf_model, c("mon = 0", "tue = 0", "wed = 0", "thu = 0")
)
```

```
##
## Linear hypothesis test:
## mon = 0
## tue = 0
## wed = 0
## thu = 0
##
## Model 1: restricted model
## Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed
##
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1     108 12.408
## 2     104 12.115  4    0.29256 0.6279 0.6437
```

joint p-value = 0.6437 > 0.05, reject H_0 : MON = TUE = WED = THU = 0

Therefore, the weekday indicator variables are not jointly significant in the reduced-form equation for $\ln(\text{PRICE})$.

This suggests that MON, TUE, WED, and THU do not have sufficient explanatory power for $\ln(\text{PRICE})$. Since these variables would be used as instruments when estimating the supply equation, they appear to be weak instruments. Therefore, we are not able to estimate the supply equation by 2SLS with confidence.